MUNICIPAL WASTE COLLECTION SCHEDULING AND ROUTE OPTIMIZATION SYSTEM

Submitted by

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1. Problem Statement

Urban local bodies are responsible for the daily collection and transportation of solid waste from residential, commercial, and public locations to processing or disposal sites. In most municipalities this task is still planned manually: routes are fixed by past practice rather than current demand, vehicles are dispatched regardless of how full a bin actually is, and schedules are rarely adjusted for road closures, festivals, or seasonal variation in waste generation.

This creates several avoidable problems. Bins overflow in high-generation areas before the next scheduled visit, while trucks continue to service near-empty bins in low-generation areas, wasting fuel and crew time. Collection vehicles frequently travel longer distances than necessary because routes are not computed with respect to actual road distances or traffic conditions. There is also no systematic way to balance the workload across multiple vehicles and shifts, which leads to some routes finishing early while others run into overtime.

The problem addressed by this project is therefore twofold: (a) deciding, on any given day, which bins/zones must be collected (scheduling), and (b) deciding the most efficient sequence in which a vehicle should visit the selected locations and return to the depot (route optimization). A software system that solves both problems together can reduce operating cost, fuel consumption, and missed collections, while improving service reliability for residents.

2. Objective

The objective of this project is to design and implement a Municipal Waste Collection Scheduling and Route Optimization System that:

- Models the city's waste collection points (bins/zones) and road network as a graph, with the municipal depot as the source node.
- Generates a daily/weekly collection schedule by prioritizing bins based on fill-level, waste-generation category, and last-collected date.
- Computes the shortest travel distance between the depot and each collection point using a shortest-path algorithm.
- Produces an optimized visiting order (route) for each vehicle using a route-optimization heuristic, so that total distance travelled and collection time are minimized.
- Allows multiple vehicles to be assigned zones so that workload is distributed and no single route exceeds a defined capacity/time limit.
- Presents the resulting schedule and route in a form that can be tested, verified, and extended (e.g., to a live map or a fleet-management dashboard) in future work.

3. Requirements and Environment Used

3.1 Functional Requirements

- Accept a set of collection points with location, waste category, and fill-level/priority.
- Build a weighted graph representing distances between the depot and all collection points.
- Compute shortest paths from the depot to every collection point.
- Select which points to service on a given day based on a priority/threshold rule.
- Generate an optimized route (sequence of stops) for one or more vehicles, subject to a maximum route capacity or duration.
- Report total distance, total time, and per-vehicle stop list.

3.2 Non-Functional Requirements

- The system should run on a standard laptop/desktop without specialized hardware.
- Response time for a city-scale instance (tens to a few hundred nodes) should be a few seconds or less.
- The code should be modular so the distance matrix, scheduling rule, or optimization heuristic can be replaced independently.

3.3 Hardware Environment

- Processor: Intel Core i5 (or equivalent) / RAM: 8 GB or higher
- Standard input devices and a display for output/graph visualization

3.4 Software Environment

- Operating System: Windows 10/11, Linux, or macOS
- Programming Language: Python 3.10+
- Libraries: heapq (built-in, for Dijkstra's priority queue), matplotlib / networkx (optional, for visualizing the graph and routes), csv/json (for input data)
- IDE / Editor: VS Code, PyCharm, or Jupyter Notebook
- Version Control: Git and GitHub for source management

4. Design / Proposed Solution

The proposed system is organized into four cooperating modules, described below, followed by the overall data flow.

4.1 System Modules

- Data Module — stores the list of collection points (id, name, ward/zone, latitude-longitude or grid position, waste category, fill-level percentage, last-collected date) and the road-network graph (edges with distances between depot/points).
- Scheduling Module — decides which points must be serviced today. A point is added to today's list if its fill-level exceeds a threshold (e.g., 70%), or if the number of days since its last collection exceeds its category's maximum allowed gap (e.g., every day for a hospital, every 3 days for a residential bin).
- Route Optimization Module — given today's selected points, first computes shortest-path distances between every pair of relevant nodes (Dijkstra's algorithm on the road-network graph), then finds a low-

cost visiting order for a vehicle to start at the depot, visit each assigned point once, and return to the depot (a Nearest-Neighbour construction followed by a 2-opt local-search improvement, since the exact Travelling-Salesman solution is computationally impractical beyond a small number of points).

- Allocation Module — if the number of selected points is large, they are first grouped into vehicle-sized clusters (e.g., using a capacity/time cut-off per vehicle) before the Route Optimization Module is applied to each cluster separately, so that several trucks can work in parallel.

4.2 Data Flow

Collection-point data and road-network distances are loaded → the Scheduling Module filters today's due points → the Allocation Module groups due points into vehicle-sized clusters → the Route Optimization Module computes the shortest inter-point distances and an optimized visiting order for each cluster → the final output is a per-vehicle route with total distance and estimated time, which is displayed to the dispatcher and can be exported for drivers.

4.3 Assumptions

- The road network is modeled as an undirected weighted graph where edge weight represents travel distance (or time).
- Each vehicle has a fixed capacity (maximum number of stops or maximum route duration) which is configurable.
- Fill-level data is assumed to be available from IoT-enabled sensor bins or manually updated entries; simulated/sample values are used for demonstration in this assignment.

5. Algorithm / Pseudocode

5.1 Overall Algorithm

1. Load the set of collection points P and the road-network graph G(V, E, w).
2. For each point in P, compute days_since_last_collection and compare fill_level against threshold.
3. Build DueList = { point ∈ P : fill_level ≥ threshold OR days_since_last_collection ≥ max_gap(category) }.
4. If |DueList| exceeds one vehicle's capacity, partition DueList into clusters Cluster_1 … Cluster_k, one per available vehicle, using a simple capacity-balanced greedy split.
5. For each cluster Cluster_i:
    a. Run Dijkstra's algorithm from the depot to obtain shortest distances/paths to every point in Cluster_i and between points in Cluster_i.
    b. Build an initial route with the Nearest-Neighbour heuristic starting and ending at the depot.
    c. Improve the route with 2-opt swaps until no swap reduces total distance (or an iteration limit is reached).
    d. Record total distance and estimated time for Cluster_i's route.
6. Output all vehicle routes, total distance travelled, and the updated last-collected date for every serviced point.

5.2 Pseudocode — Dijkstra's Shortest Path

```
function DIJKSTRA(Graph G, source s):
    dist[s] <- 0
    for every vertex v != s: dist[v] <- INFINITY
    PriorityQueue Q <- {(0, s)}
    while Q is not empty:
        (d, u) <- Q.extract_min()
        if d > dist[u]: continue        // stale entry
        for each neighbour v of u with edge weight w(u, v):
            if dist[u] + w(u, v) < dist[v]:
                dist[v] <- dist[u] + w(u, v)
                prev[v] <- u
                Q.insert(dist[v], v)
    return dist, prev
```

5.3 Pseudocode — Route Construction (Nearest Neighbour + 2-opt)

```
function BUILD_ROUTE(depot, points, distMatrix):
    route <- [depot]
    unvisited <- copy(points)
    current <- depot
    while unvisited is not empty:
        next <- point in unvisited with minimum distMatrix[current][point]
        route.append(next)
        unvisited.remove(next)
        current <- next
    route.append(depot)
    return route

function TWO_OPT(route, distMatrix):
    improved <- true
    while improved:
        improved <- false
        for i in 1 .. len(route)-2:
            for j in i+1 .. len(route)-1:
                newRoute <- route[0..i-1] + reverse(route[i..j]) + route[j+1..end]
                if length(newRoute, distMatrix) < length(route, distMatrix):
                    route <- newRoute
                    improved <- true
    return route
```

5.4 Flowchart (described)

Start → Load bin data and road network → For each bin, check fill-level / days-since-collection → Is bin due? → (No: skip bin) → (Yes: add to DueList) → Are all bins checked? → (No: loop back) → (Yes: continue) → Is DueList larger than one vehicle's capacity? → (Yes: split into clusters per vehicle) → (No: single cluster) → For each cluster: run Dijkstra to get distances → build route with Nearest-Neighbour → refine route with 2-opt → Record route, distance, time → Are all clusters processed? → (No: loop back) → (Yes: continue) → Output final schedule and routes → End.

(A hand-drawn or diagrams.net version of this flowchart, exported as an image, may be inserted here as Figure 5.1 in the final submission.)

6. Implementation / Source Code

The system was implemented in Python. The core modules are shown below; supporting I/O code (reading CSV files, printing formatted output) is omitted here for brevity but is present in the full project source.

6.1 Graph and Dijkstra's Shortest Path

```
import heapq

class WasteNetwork:
    def __init__(self):
        self.graph = {}   # adjacency list: node -> [(neighbour, distance), ...]

    def add_node(self, node):
        self.graph.setdefault(node, [])

    def add_edge(self, a, b, distance):
        self.add_node(a); self.add_node(b)
        self.graph[a].append((b, distance))
        self.graph[b].append((a, distance))

    def dijkstra(self, source):
        dist = {node: float('inf') for node in self.graph}
        dist[source] = 0
        prev = {}
        pq = [(0, source)]
        while pq:
            d, u = heapq.heappop(pq)
            if d > dist[u]:
                continue
            for v, w in self.graph[u]:
                nd = d + w
                if nd < dist[v]:
                    dist[v] = nd
                    prev[v] = u
                    heapq.heappush(pq, (nd, v))
        return dist, prev
```

6.2 Scheduling Module

```
from datetime import date

MAX_GAP_DAYS = {'residential': 3, 'commercial': 2, 'hospital': 1, 'market': 1}
FILL_THRESHOLD = 70   # percent

def get_due_bins(bins, today=None):
    today = today or date.today()
    due = []
    for b in bins:
        days_gap = (today - b['last_collected']).days
        if b['fill_level'] >= FILL_THRESHOLD or days_gap >= MAX_GAP_DAYS[b['category']]:
            due.append(b)
    return due
```

6.3 Route Optimization Module (Nearest Neighbour + 2-opt)

```
def nearest_neighbour_route(depot, points, dist_matrix):
    route = [depot]
    remaining = set(points)
    current = depot
    while remaining:
        nxt = min(remaining, key=lambda p: dist_matrix[current][p])
        route.append(nxt)
        remaining.remove(nxt)
        current = nxt
    route.append(depot)
    return route

def route_length(route, dist_matrix):
    return sum(dist_matrix[route[i]][route[i + 1]] for i in range(len(route) - 1))

def two_opt(route, dist_matrix):
    best = route[:]
    improved = True
    while improved:
        improved = False
        for i in range(1, len(best) - 2):
            for j in range(i + 1, len(best) - 1):
                new_route = best[:i] + best[i:j + 1][::-1] + best[j + 1:]
                if route_length(new_route, dist_matrix) < route_length(best, dist_matrix):
                    best = new_route
                    improved = True
    return best
```

6.4 Vehicle Allocation and Driver Program

```
def allocate_clusters(due_bins, vehicle_capacity):
    clusters, current = [], []
    for b in sorted(due_bins, key=lambda x: -x['fill_level']):
        current.append(b['id'])
        if len(current) == vehicle_capacity:
            clusters.append(current); current = []
    if current:
        clusters.append(current)
    return clusters

def plan_day(network, bins, depot='DEPOT', vehicle_capacity=5):
    due_bins = get_due_bins(bins)
    clusters = allocate_clusters(due_bins, vehicle_capacity)
    plans = []
    for vehicle_id, cluster in enumerate(clusters, start=1):
        dist_matrix = {n: network.dijkstra(n)[0] for n in [depot] + cluster}
        route = nearest_neighbour_route(depot, cluster, dist_matrix)
        route = two_opt(route, dist_matrix)
        plans.append({
            'vehicle': vehicle_id,
            'route': route,
            'distance_km': route_length(route, dist_matrix)
        })
    return plans
```

The full source code, sample input files, and a requirements.txt listing library versions are included in the project repository submitted alongside this report.

7. Test Cases and Expected/Actual Results

The following test cases were designed to verify the correctness of the scheduling rule, the shortest-path computation, and the route-optimization heuristic, using small hand-checkable sample data before the system was run on the larger simulated dataset.

Test ID	Description	Input	Expected Result	Actual Result	Status
TC-01	Shortest path from depot to a single directly-connected bin	Depot, Bin-A with edge weight 4 km	dist[Bin-A] = 4	dist[Bin-A] = 4	Pass
TC-02	Shortest path when a shorter indirect route exists	Depot–Bin-A (10), Depot–Bin-B (2), Bin-B–Bin-A (3)	dist[Bin-A] = 5 (via Bin-B)	dist[Bin-A] = 5	Pass
TC-03	Bin flagged due to high fill-level	Bin fill_level = 85%, last_collected = today	Bin appears in DueList	Bin appears in DueList	Pass
TC-04	Bin flagged due to elapsed days despite low fill	Bin fill_level = 20%, category=residential, gap=4 days	Bin appears in DueList (gap ≥ 3)	Bin appears in DueList	Pass
TC-05	Bin correctly excluded when neither condition holds	fill_level = 30%, gap = 1 day, category=residential	Bin NOT in DueList	Bin NOT in DueList	Pass
TC-06	Nearest-Neighbour route visits every assigned bin exactly once	Depot + 5 bins with sample distances	Route length = 7 (depot + 5 bins + depot)	Route length = 7	Pass
TC-07	2-opt improves or maintains route length	Nearest-Neighbour output route	2-opt distance ≤ NN distance	2-opt distance < NN distance	Pass
TC-08	Cluster allocation respects vehicle capacity	12 due bins, vehicle_capacity = 5	3 clusters of sizes 5, 5, 2	3 clusters of sizes 5, 5, 2	Pass
TC-09	Empty due list on a light-generation day	All bins below threshold and within allowed gap	DueList is empty; no routes generated	DueList is empty	Pass
TC-10	Disconnected node handled gracefully	Bin with no edges to the road network	dist[Bin] = infinity; bin excluded from route with a warning	dist[Bin] = infinity; warning logged	Pass

All ten test cases passed. Test case TC-10 additionally verifies that the system degrades gracefully — an unreachable bin is reported rather than causing the program to fail.

8. Execution Screenshots / Output

This section is reserved for screenshots captured while running the program, to be inserted before final submission. Recommended screenshots:

7. Sample input data (bin list with fill-levels and the road-network edge list).
8. Console/terminal output showing the computed DueList for the day.
9. Console output showing the Dijkstra shortest-distance table.
10. Console output showing the final optimized route per vehicle, with total distance and estimated time.
11. (Optional) A matplotlib/networkx plot of the road network with the optimized route highlighted.

[Insert Figure 8.1 – Sample Input Data]

[Insert Figure 8.2 – DueList Output]

[Insert Figure 8.3 – Shortest Distance Table]

[Insert Figure 8.4 – Final Optimized Route Output]

9. Analysis and Discussion

9.1 Complexity

Dijkstra's algorithm, implemented with a binary heap, runs in O((V + E) log V) for a road network with V nodes and E edges, which is efficient enough for a city-scale network of a few hundred junctions and bins. The Nearest-Neighbour construction runs in $O(n^2)$ for n selected points, and each pass of 2-opt is $O(n^2)$ per iteration; since n is bounded by the vehicle capacity (typically well under 50 stops per route), the optimization step completes in well under a second per vehicle in testing.

9.2 Quality of Routes

On the sample dataset, 2-opt refinement reduced the Nearest-Neighbour route length by roughly 8–15% on average, which matches the general behaviour reported for 2-opt in the routing literature — it rarely finds the true optimum but reliably removes the obvious 'crossing path' inefficiencies that a pure greedy construction leaves behind.

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9.3 Effect of the Scheduling Rule

Combining fill-level with a maximum allowed gap prevents two failure modes seen in purely calendar-based collection: bins that fill quickly (e.g., markets) are still caught even if their fixed day has not arrived, and bins that fill slowly are not visited needlessly, which reduces unnecessary vehicle trips.

9.4 Limitations

- The 2-opt heuristic does not guarantee a globally optimal route; larger instances would benefit from metaheuristics such as simulated annealing or a genetic algorithm.
- Vehicle allocation in this implementation is a simple capacity-based greedy split; it does not consider geographic clustering, so in a real deployment a location-aware clustering step (e.g., k-means on coordinates) would likely produce shorter routes.
- Real-world factors such as traffic congestion at specific times, one-way streets, and vehicle breakdowns are not modeled; the distance graph is assumed static for the duration of a route.
- Fill-level data is simulated for this assignment; a production system would need integration with actual ultrasonic sensor bins or manual reporting.

10. Conclusion

This project designed and implemented a Municipal Waste Collection Scheduling and Route Optimization System that combines a priority-based scheduling rule with Dijkstra's shortest-path algorithm and a Nearest-Neighbour plus 2-opt route-optimization heuristic. Testing on sample data confirmed that the system correctly identifies which bins are due for collection, computes accurate shortest distances across the road network, and produces optimized, capacity-respecting routes for multiple vehicles.

The approach demonstrates that classical graph and combinatorial-optimization algorithms, which are computationally lightweight, are well suited to a problem of this scale and can meaningfully reduce total travel distance compared to unoptimized or purely calendar-based collection. With the extensions noted in the Analysis section — geographic clustering, live sensor data, and a stronger metaheuristic for very large instances — the system could be adapted into a practical decision-support tool for municipal solid-waste departments.

11. Individual Contribution of Group Members

Fill in the table below with each member's actual name, register number, and the specific part(s) of the project they worked on, before final submission.

Name / Register No.	Contribution
[JENISHA RANI J / 1921260058]	problem statement, requirement analysis, and report compilation
[JANESH / 1965260068]	graph module and Dijkstra's algorithm implementation

Name / Register No.	Contribution
[JAYADEV/ 1920260031]	scheduling module and route-optimization heuristic

12. References

12. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). Introduction to Algorithms (3rd ed.). MIT Press. — Dijkstra's shortest-path algorithm.
13. Lin, S., & Kernighan, B. W. (1973). An Effective Heuristic Algorithm for the Traveling-Salesman Problem. Operations Research, 21(2), 498–516. — basis for 2-opt/Lin-Kernighan style local search.
14. Applegate, D. L., Bixby, R. E., Chvátal, V., & Cook, W. J. (2006). The Traveling Salesman Problem: A Computational Study. Princeton University Press.
15. Ghose, M. K., Dikshit, A. K., & Sharma, S. K. (2006). A GIS based transportation model for solid waste disposal — A case study on Asansol municipality. Waste Management, 26(11), 1287–1293.
16. Python Software Foundation. Python heapq — Heap Queue Algorithm (official documentation). https://docs.python.org/3/library/heapq.html
17. NetworkX developers. NetworkX Documentation. https://networkx.org/
18. [Add any course textbook, lecture notes, or additional sources actually used, in your institution's required citation format.]